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The role of User Entity Behavior Analytics to detect network attacks in real time

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***Abstract*— Organizations are using advanced security solutions to protect their information resources. However, even such high investments, traditional security approaches failed to protect the network structure against state-of-the-art attacks. New proactive approaches to security are on the rise such as User Entity Behavior Analytics (UEBA). UEBA is a type of cybersecurity process that uses machine learning, algorithms, and statistical analyses to detect real-time network attacks. This paper aims to assess the value and success of using behavior analytics in securing the network from not-before-seen attacks such as zero-day attacks. This paper uses a systematic literature review and self-administrated survey and interviews with convenience sampling of high profile network users and top security vendors. Survey and interviews with various security experts are utilized to verify the matter-of-fact effectiveness of the solutions based on behavior analytics. During collecting the primary data via a survey, researchers will go for a structured interview with vendors who are selling solutions to understand the performance of behavior analytics-based solutions and the distinct features of their solutions. The results of literature review, survey, interviews and focus groups will be used to assess the value and success of using behavior analytics in securing the network from not-before-seen attacks such as zero-day attacks. The endeavor of this paper is to highlight the weaknesses and strengths of different UEBA solutions and their effectiveness for detecting network attacks in real-time interaction. This research contrasts top fifteen UEBA technologies based on use cases and capabilities and highlights common usage scenarios. Based on the evidence, recommendations will be given.**

Keywords*—— *traditional security approaches, Security attacks, User Entity Behavior Analytics, Real-time

I. Introduction

Today's security threat landscape has many constantly changing forms and shapes. According to Young (2017), these changing forms of threat making it nearly impossible to be detected with traditional security approaches. Convertino (2017) reported that companies around the world are spending billions of dollars on network security and yet 90% of them are breached) [1] [2].

According to the Data Breach Investigations Report [3], analysis of 53,000 security incidents, shows that 48% of incidents was hacking and malware was second at 30%.

Most cybercriminals take few minutes or less to compromise a system, but just 3% of security breaches are discovered as they accrued, while 68% went undiscovered for months or more [4].

Most breach cases, a third party detect the data breach within the organization, like law enforcement which revealed the security breaches in TJX Companies in 2006, and VeriSign in 2010, or a partner who notified Heartland Payment Systems in 2008, and Yahoo in 2014 of the breaches. Worst of all, many breaches are detected by customers like what happened to Adobe in 2015, and LinkedIn in 2016.

Malware incorporates techniques to extract user credentials stored in the memory of the compromised system. These can include credentials of domain users or admins who log into the machine, detecting and blocking these attacks comes down to being able to identify malicious behaviors and respond fast to mitigate damage.

Malware repeatedly changes its techniques in order to avoid being discovered, according to (Data Breach Investigations Report, 2018) 37% of malware signatures appear only once. Securing a network perimeter using a signature-based approach to protect organization systems and data is no longer effective. It is important to be able to detect the abnormal behavior, credentials abuse, network misusage and isolating network devices which their security is compromised in order to deactivate attackers and respond quickly and effectively [4] [3].

The growing of adopting security architecture enriched with advanced analytics and machine learning, instead of the traditional approaches which cannot detect and prevent the recent polymorphic security threats, has decreased Malware breaches by 20% in one year [3] this significant drop brings cybersecurity to a completely new level.

Gartner Security and Risk Management Summit (2016) has addressed the urge for approaches that can detect insider threats and external hackers effectively and efficiently, therefore few new trendy approaches have been introduced such as user and entity behavioral analytics (UEBA) and anticipating security identity event management (SIEM) with user and entity behavioral analytics (UEBA) evolution.

User and Entity Behavioral Analytics (UEBA), the first product was called User Behavioral Analytics (UBA) has been on the market since 2015. Later in August of 2015, Avivah Litan's Gartner analyst introduced the term "entity" into the title creating "user and entity behavioral analytics" (UEBA), as vendors added the capability of monitoring and analyzing the behavior of entities besides the users. In 2016, Gartner Security and Risk Management Summit categorized (UEBA) as a risk management solution in use but still in the Innovation Trigger stage [5].

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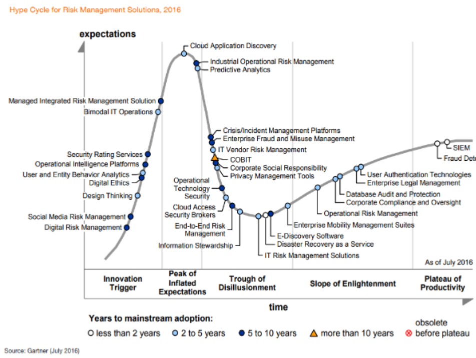

Figure .1. Hype Cycle for Risk Management Solutions

(Source; Gartner, 2016)

UEBA embed unsupervised machine learning to spot significant changes in user, device or network activity that indicates attack. Security teams can set the priority of alerts to rectify in order to mitigate security risks, moreover UEBA can utilize automated response procedures in order to reduce time-to-response cycle.

UEBA vendors offer a range of capabilities with the support of machine learning and analytic models, UEBA analyze and monitor the users behaviors and at the same time the other entities to detect the signs that indicate the attacks, use advanced analytics to detect multiple kinds of threats, and offer the ability to correlate multiple anomalous activities that could be related to a single security incident, all these capabilities are performed in real time or near-real time and offer a new kind of threat visibility.

For an emerging technology like UEBA, there is a need to asses and evaluate UEBA vendors base on the capabilities which they offer. Therefore, this study aims to use a systematic literature review and self-administrated survey and interviews to assess the value and success of using behavior analytics in securing the network from not-before-seen attacks, and to verify the effectiveness of the solutions based on behavior analytics.

The endeavor of this study is to highlight the weaknesses and strengths of different UEBA solutions and their effectiveness for detecting network attacks in real-time interaction. This study compares the top fifteen UEBA technologies base on capabilities and use cases and focuses in common scenarios of usage. Build on the evidence, recommendations will be given.

II. Current cybersecurity Approaches

Today, signature-based detection is still the predominant method of finding and eliminating security threats in the enterprise, Intrusion-prevention systems and antivirus software only look to match attack fingerprints They check the activities and recognize that one which matches the signature of a known attack then suspicious activity will be prevented.

Attackers use sophisticated methods to skip systems such as Intrusion Prevention System and Intrusion Detection System IPS/IDS by using techniques like Denial of Service, fragmentation, obfuscation, and application seizing, they try to pass their attack as a legitimate traffic to prevent IPS/IDS from detecting the security breach.

Some IPS/IDS depend on abnormality detection; by comparing the current traffic with the baseline traffic, alert will show when unfamiliar traffic is shown. IPS/IDS based on abnormality detection have a more accurate and customizable intrusion detection methods but from the administrative side it is more intensive and need more computational expense.

It is universally accepted that IPS/IDS are insufficient to point out all attacks especially zero-day attacks, repeated false positive alerts, and incompetence to provide a worthy Return on Investment (ROI).

While the shortcomings of signatures used by IPS/IDS and other perimeter security systems are well known, much of the industry effort has been focused on delivering signatures faster. This tactic has simply led to faster attackers or the use of vectors with unknown signature.

In order to protect organization network from these unknown threats and attacks, many Next-Gen security solutions are using a variety of emerging approaches and technologies, which are mostly relying on some form of advanced analytics to monitor network behavior and stop unusual events.

III. Literature review

In an article on "The Expanding Role of Data Analytics in Threat Detection" by SANS Institute, Barbara Filkins (2015) argues that most systems depend on one or more of three threat detection methodologies - Signature-Based (or Misuse) Detection, Anomaly-Based (or Behavior-Based) Threat Detection, and Continuous System Health Monitoring [6].

The first method, Signature-Based (or Misuse) Detection uses a set of rules to identify threats such as intrusions and viruses by watching for patterns of events specific to known and documented attacks. Preprocessing methods, such as deep packet inspection for network traffic, find possible signatures in captured network traffic. The resulting signatures from the monitored environment are matched to known signatures in a signature database. alert is issued in case if any match appear and the detector will do nothing in case of there is no match at all.

This method typically produces fewer false positives than traditional methods, with relatively low processing demands. The main disadvantage is that it only detects attacks for which a defined signature is known and available. New attacks must be identified, modeled and added to signature databases, which must be updated regularly to keep innovative exploits, or those based on previously unknown flaws, from evading defenses.

The second method, Filkins (2015) states that an Anomaly-Based or (Behavior-Based) in which threat detection depends on the assumption that attack behaviors differ enough from normal activity that malicious actions can be detected and identified. Tools using this method begin by creating models of behavior patterns that represent "normal" behavior for the networks, systems, applications, end users and devices that make up the environment in which they're installed. They then look for deviations from that pattern.

The second method advantage is the ability of it to discover the threat without knowing the significance of it. Historically, this method has high false positive rates, the complicity of training a system in a highly dynamic environment, and computational expense.

The third method, Continuous System Health Monitoring detects intrusions by active monitoring of key system "health" or performance factors to identify suspicious changes or trends in activity and resource usage. On the network, this may mean monitoring the protocols used in the network, bandwidth usage over time, consider ports which has unexpected traffic increases

This method maps both to unique behaviors that malicious activities may have in common by using the developed tune system-wide measures and understand the significance of identified variations and trends.

A security platform uses a set of mechanisms and techniques to detect any anomalousness behavior and security breaches within the network. The security platform apply "big data" techniques and perform security analytics by utilizing machine learning.

The security platform performs 'real-time' user/entity behavioral analytics (UEBA) to detect the security-related anomalies and threats, whether such suspicious behavior were previously known or not. A visual presentation for the analytical results shows the rank of risk with the supporting evidence, this visual presentation enables network security administrators to response and take action to detected anomaly or threat immediately. Network administrators can spot abnormal behavior by searching the behaviors' patterns [7].

IV. UEBA Design

UEBA work to detects the changes in the communication's behavior between server and endpoint which may indicate an attack. By using unsupervised machine learning that works to detect considerable changes in device, user or network activity that signal the presence of an attack. security administrators can rectify the high priority alerts first in order to protect sensitive assets, or they can choose automated response procedures to minimize time-to-contain cycle.

Figure 2 shows that behavior analytics process is generally powered by machine learning which apply mathematical models to view the analytics of user, entity and network's behavior. hence this approach allows the detecting of existing attacks within the enterprise.

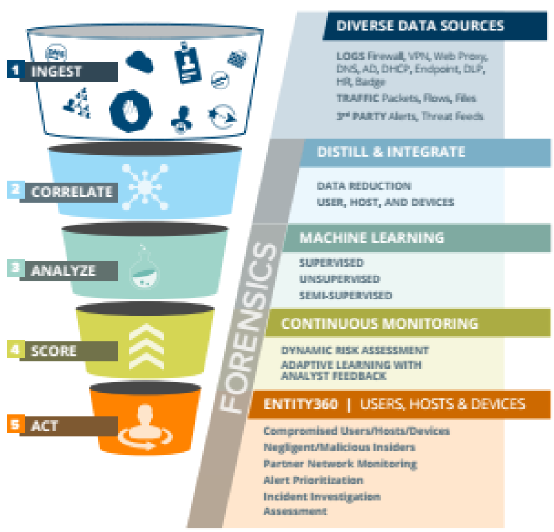

Figure.2. User Entity Behavior Analytics Model

(Source; ARUBA, 2018)

This approach is being used successfully by some of the network security vendors who claim that WanaCry or Petya Ransomware could not hack the clients on their watch [8]. Therefore, network security vendors suggest enterprises deploy their solutions based on behavioral analytics to detect intrusions that slip traditional preventive technologies.

Although vendors add different features and keep their methods for anomaly detection secret, there are some basic characteristics common to all UEBA tools. UEBA monitors and collects data of activities of user and entity from system logs and analyses the collected data using advanced analytical methods. A baseline profile of user behavior and entity is created by finding behavioral patterns and deviations from acceptable or normal activity and thresholds of normal behavior are established. Creating a baseline profile is the most important stage of a UEBA system, as it defines the accuracy of further detection of potential.

Initially, profile construction is done using training data from the application data repository. Baseline modeling of user and entity behavior profile is provided by UEBA vendor. This profile is the most critical asset in any system because reports and alerts are based on the accuracy of this profile. A raw profile is created and stored in the profile repository. Then the Profile application which is provided by the vendor performs profile operation on this raw profile and sends it to profile evaluation which generates evaluation results, this evaluation results to be used by profile construction to enhance the quality of profiles. Profile construction is continually updated either by results of profile evaluation or testing dataset (stored in the application data repository).

To improve the accuracy of behavior profile in order to support its applications effectively and efficiently, it is essential to automate or semi-automate profiling systems throughout the process of constructing, evaluating and storing profiles, and providing profile operations for developing profiling applications [9].

User behavior profiling models include among others machine learning, statistical models, rules & signatures based models and out of the box analytics models for different use cases. The effectiveness of the aforementioned models are measured by the capability of the model in base-lining and profiling dynamically so that it quickly adapts to user role changes etc.

Standalone UBEA tools build profiles/models for the behavior of Users and Entities over a period of time and use that as a baseline to detect any malicious actions by noting any abrupt or sudden change in their behavior. The current user and entity behavior is compared with the established baseline and then UEBA system decides whether deviations are acceptable or anomalous. In case an anomaly is detected, the system estimates the degree of deviation and its risk level and sends alerts to security officers in real time.

Zero-day attacks and Ransomware are two big security threats that have caused enormous damage to enterprises in recent years. Financial losses are reaching millions because it is not just the cost of the ransom but many other associated costs. In order to protect organization network from these attacks, many Next-Gen Security Solutions are using a variety of emerging approaches and technologies, mostly relying on some form of advanced analytics to monitor network behavior and stop unusual events.

V. THE SIGNIFICANCE OF UEBA APPROACH

Solutions based on behavior analytics maintain a database of system objects and also store the system's normal and abnormal behavior called anomaly detection. Anomaly detection attempts to determine if a data pattern does not conform to expected normal behavior. A straightforward approach, therefore, is to define what is considered as a normal behavior and state any observation in the data which is not a normal behavior as an anomaly [10]. Recently, many techniques have been developed in past years for spotting outliers and anomalies [11]. Soft-computing and machine learning techniques are rigorously used to build autonomous anomaly detection.

Singh and Nene (2013) made a survey on machine learning techniques for an anomaly. Pattern matching, string matching, and machine learning techniques and hardware computation power are used to provide a great and strongest security against numerous attacks detection [12]. "It is important to design big data-driven cybersecurity solutions that effectively and efficiently derive actionable intelligence from available heterogeneous sources of information using principled data analytic methods" to mitigate cyber threats [13]Authors further added that the data analytics need to integrate various advanced machine learning algorithms and human driven knowledge for iteration aggregated learning. looking at the need for analyzing huge amount of collected data in real-time processing, a scalable data processing supporting structure is required to handle big data with low latency.

Tools based on data science and machine learning can help organizations quickly detect malicious activity and act according to the inherent risk presented by potential rogue elements. Monitoring the critical performance characteristics, such as network activity, computer's processor are obtainable by using the analytical methods. Furthermore, analytics are used to spot abnormal behavior of users, entities and applications inside the organization, this done by establishing normal baseline behavior over a period of time, then draw a behavioral model and compare the registered real-time activity against this model to flag abnormal behavior [6].

According to Ernst & Young Analytics Survey (2016), successful and secure organizations are harnessing more sophisticated analytics tools and data visualization to identify patterns and trends of rogue activities [14]. The survey found that in order to enhance information security and reduce vulnerability, firms are implementing close observation systems corroborative by powerful data analytics to arrange the various data units and manage growing regulatory compliance. In order to keep continuous control of firm activity, communications, and critical infrastructure, IT security leaders are investing in programs that pattern and observe behavioral models.

Companies transitioned from current reactive surveillance activities into proactive surveillance where user behavioral analytics capabilities are used as a basis for user behavior patterns and anomalies.

With thousands of integrated analytical capabilities and algorithms for natural language processing, pattern discovery/detection, and ongoing surveillance, behavior-based analytics solutions can automatically alert organizations to potential policy breaches and significant negative trends or events. Because of its perpetual machine learning capability, behavior analytics understands which activities commonly occur in any particular scenario, flagging for human attention objects or behaviors that are out of the ordinary.

Commenting on the ranking of top information security technologies released at a security conference by Gartner Inc., Leopold [15] argues that a user and entity behavioral analytics (UEBA) is emerging as a leading information security technology for embattled IT managers. Releasing its list of top information security technologies Gartner [16] argues that (UEBA) is become known as a most influential information security technology for IT security leaders. Releasing its list of top information security technologies Gartner [16] argues that UEBA gives analytics centralized not only around user behavior, but around other entities (endpoints, networks, and applications) also. This correlation of the analyses cover several entities ensure more accurate results and more effective threat detection.

Gartner recommends using UEBA (User Entity Behavior Analytics) coupled with SIEM (Security Information & Event Management along with EDR (Endpoint Detection and Response). Since UEBA products typically need a data source, and SIEM products are commonly the central aggregation point for security logs for an organization, these tools complement each other. It can almost be a two-way street in the sense that the SIEM forwards applicable log events to the UEBA for user profiling, while the UEBA tool generates alerts that can be sent into the SIEM tool for enhancement of the events from other sources, in turn presenting the alert to a security analyst for triage. Also, a SIEM solution in place with the necessary data makes the UEBA deployment far easier, as data collection becomes straightforward. This is the reason that we have recently seen some of the UEBA vendors building their own SIEM solutions and competing in that market as well. Artificial Intelligence (AI), Machine Learning (ML), Neural networks, deep neural networks, Hidden Markov Model (HMM), and Support Vector Machines (SVM) only define the inner workings of solutions based on UEBA. This research is limited to contrasting select UEBA technologies derive from use cases and capabilities and points out common usage scenarios and tool evaluation processes.

VI. UEBA DRAWBACKS

The negative part of applying machine learning in UEBA is the same drawbacks that any machine learning brings. Machine learning has limitations dealing with privileged users, developers, and knowledgeable insiders. Those users represent a unique situation because their job functions often require irregular behaviours. This cause difficulties for statistical analysis to create a baseline the algorithms. Another drawback is that UEBA can't indicate the long-term sophisticated "low and slow" as attacks because they does not have day to day impact and become as if non-existent.

VII. DISCUSSION

Zero-day attacks and Ransomware are two big security threats that have caused enormous damage to enterprises in recent years. Financial losses are reaching millions because it is not just the cost of the ransom but many other associated costs. In order to protect organization network from these attacks, many Next-Gen Security Solutions are using a variety of emerging approaches and technologies, mostly relying on some form of advanced analytics to monitor network behavior and stop unusual events.

Protection in all these solutions is not created equal. Some solutions simply aren't as effective at stopping an attack. Users are simply indecisive which solution is best for them. This thesis will try to deliver powerful insights into the prevalent security issues facing organizations today, by evaluating the strength and weakness of the UEBA approach, identifying a typical model of UEBA with comprehensive detection features and identifying a UEBA solution that will provide most reliable protection.

VIII. CONCLUSION

UEBA employs machine learning to build profiles/models for the behavior of Users and Entities over a period of time, by applying security analytics UEBA can detect behavior's anomalies and threats in a computer network environment. It is detecting security-related anomalies and threats in both real-time and batch paths/modes. UEBA provides visual presentation for the analytical results shows the rank of risk with the supporting evidence, this visual presentation enables network security administrators to response and take action to detected anomaly or threat immediately. This paper provides an overview of signature-based approaches and –their limitation. Then it expands the overview over UEBA and the related research work and then describes the basic characteristics of UEBA design which are common to all UEBA tools. Then this paper evaluates and presents UEBA approach both significances and drawbacks. The researchers will enhance the understanding of UEBA approach with contrasts top UEBA technologies and validated them using real-world datasets.

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